

MuTau Channel MC Normalization Documentation

Author: Analysis Team **Date:** February 2026 **Analysis:** Exclusive Di-Tau Production ($\gamma\gamma \rightarrow \tau\tau$)

1. Overview

This document describes the Monte Carlo (MC) normalization procedure for the MuTau channel analysis. The goal is to properly weight MC events so that the expected number of events matches:

$$N_{\text{expected}} = \text{Luminosity} \times \text{Cross-section} \times \text{Efficiency} \times \text{Branching_Ratio}$$

2. Samples Used

2.1 Data

- **Dataset:** 2018 Ultra Legacy (UL)
- **Integrated Luminosity:** 54.9 fb^{-1} ($54,900 \text{ pb}^{-1}$)
- **File:** Data_2018_UL_MuTau_nano_merged_proton_vars.root

2.2 Drell-Yan (DY)

- **Process:** $Z/\gamma^* \rightarrow \ell\ell$ ($M > 50 \text{ GeV}$)
- **Generator:** aMC@NLO + Pythia8
- **Dataset:** /DYJetsToLL_M-50_TuneCP5_13TeV-amcatnloFXFX-pythia8/RunIISummer20UL18NanoAODv9
- **Cross-section:** 6077.22 pb (NNLO, FEWZ 3.1)
- **File:** DY_2018_UL_MuTau_nano_merged_pileup_protons.root

2.3 ttbar

- **Process:** $t\bar{t} \rightarrow \text{all decays}$
- **Generator:** aMC@NLO + Pythia8
- **Dataset:** /TTJets_TuneCP5_13TeV-amcatnloFXFX-pythia8/RunIISummer20UL18NanoAODv9-106X_upg
- **Cross-section:** 831.76 pb (NNLO+NNLL, TOP++ 2.0)
- **File:** ttjets_2018_UL_MuTau_nano_merged_pileup_protons.root

2.4 QCD Multijet

- **File:** QCD_2018_UL_MuTau_nano_merged_proton_vars_new.root
- **Note:** Data-driven estimation, no MC normalization needed

2.5 Signal ($\rightarrow$)

- **File:** MuTau_sinal_SM_2018_july.root
 - **Note:** Fully normalized in weight branch
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3. Normalization Formula

The correct per-event weight for MC is:

$$\text{weight} = (\mathbf{L} \times \mathbf{sigma} / \mathbf{Sum_w_gen}) \times \mathbf{generator_weight} \times \mathbf{SF_muon} \times \mathbf{eff_proton}$$

Where: - **L** = Integrated luminosity = $54,900 \text{ pb}^{-1}$ - **sigma** = Production cross-section (pb) - **Sum_w_gen** = Sum of generator weights from original NanoAOD - **generator_weight** = Per-event generator weight (from aMC@NLO) - **SF_muon** = Muon scale factors (trigger $\times$ ID/Iso $\times$ reconstruction) - **eff_proton** = Proton acceptance = 0.245 (probability of protons on both PPS arms)

4. Sum of Generator Weights

Extracted from the `Runs` tree in original NanoAOD files using `genEventSumw`:

Sample	N_events	$\Sigma(\text{generator_weights})$
DY	195,510,810	3.323×10^{12}
ttbar	304,895,029	6.108×10^{11}

Script used: `check_mc_normalization.py`

5. Scale Factors

5.1 MC Scale Factors

$$\text{Scale} = (\mathbf{L} \times \mathbf{sigma}) / \mathbf{Sum_w_gen}$$

Sample	Calculation	Scale Factor
DY	$(54900 \times 6077.22) / 3.323 \times 10^{12}$	1.004×10
ttbar	$(54900 \times 831.76) / 6.108 \times 10^{11}$	7.476×10

5.2 Proton Acceptance

`eff_proton = 0.245`

This represents the probability that random pileup protons are reconstructed on both PPS arms (arm 0 AND arm 1).

6. Processing Chain

6.1 Phase 0 (Skimming)

Scripts: `fase0_dy.py`, `fase0_ttjets.py`

- Reads NanoAOD files
- Applies basic preselection
- Extracts `generator_weight = Generator_weight` from NanoAOD

6.2 Phase 1 (Selection)

Scripts: `fase1_dy.py`, `fase1_ttjets.py`

Applies analysis cuts: - Muon ID 3 (tight) - Tau ID: $VS_{jet} > 63$, $VSe > 7$, $VS_{mu} > 1$ - $\Delta R(,) > 0.4$ - $|| < 2.4$ for both leptons - $pT() > 35$ GeV, $pT() > 100$ GeV - Opposite-sign requirement

Calculates event weight:

DY (fase1_dy.py line 226):

```
event_weight = generator_weight * mu_trig_sf * mu_idiso_sf * mu_reco_sf
```

ttbar (fase1_ttjets.py line 217):

```
event_weight = 0.15 * mu_trig_sf * mu_idiso_sf * mu_reco_sf
```

Note: ttbar uses a hardcoded 0.15 factor instead of `generator_weight`!

6.3 Merging with Proton Mixing

Script: `merge_pp_mutau.py`

- Merges phase 1 output files
- Mixes random pileup protons from PPS
- Applies proton acceptance:

```
weight = event_weight * 0.245
```

7. Weight Content in Final Files

7.1 DY Weight Branch

```
weight = generator_weight * muon_SF * 0.245
        17000 * 0.97 * 0.245
        4000 (average)
```

7.2 ttbar Weight Branch

```
weight = 0.15 * muon_SF * 0.245
        0.15 * 0.97 * 0.245
        0.036 (average)
```

8. Plotting Stage Normalization

8.1 DY

The weight branch contains $\text{generator_weight} \times \text{SFs} \times 0.245$, so we need to apply the MC scale factor:

```
const double DY_SCALE = 1.004e-4; // = L*sigma/Sum_w

// In TTree::Draw
TString cut_dy = "(selection) * weight";
t_dy->Draw("var >> h_dy", cut_dy, "goff");

// After filling
h_dy->Scale(DY_SCALE);
```

Final per-event weight:

```
w_final = weight * DY_SCALE
        = (gen_w * SFs * 0.245) * 1.004e-4
        17000 * 0.97 * 0.245 * 1.004e-4
        0.4
```

8.2 ttbar

The weight branch already contains the correct normalization via the 0.15 factor:

```
const double TTBAR_SCALE = 1.0; // No additional scaling needed

// In TTree::Draw
TString cut_ttbar = "(selection) * weight";
t_ttbar->Draw("var >> h_ttbar", cut_ttbar, "goff");
```

```
// After filling
h_ttbar->Scale(TTBAR_SCALE);
```

Why 0.15 works:

```
Theoretical factor = L*sigma/Sum_w * avg_gen_w
                   = 7.476e-5 * 2000
                   = 0.1495  0.15
```

9. Validation

9.1 Expected Yields After Proton Cut

From `validate_normalization.py`:

Sample	Raw Events	Sum of Weights	After Scale	Final Yield
DY	1,704	4,836,853	$\times 1.004e-4$	486 events
ttbar	26,649	954	$\times 1.0$	954 events

9.2 Cross-Check: ttbar 0.15 Factor

```
Theoretical: L*sigma/Sum_w * avg_gen_w = 7.476e-5 * 2000 = 0.1495
Empirical:   0.15
Ratio:       0.15 / 0.1495 = 1.003
```

The empirical 0.15 factor matches the theoretical calculation to within 0.3%.

10. Code Implementation

10.1 MuTauPlots__after.C (Updated)

```
// Configuration
const double DY_SCALE = 1.004e-4;           // L*sigma/Sum_w for DY
const double TTBAR_SCALE = 1.0;             // ttbar already normalized

// Proton requirements for MC (pileup protons)
TString proton_cut_mc = "xi_arm1_1 >= 0 && xi_arm2_1 >= 0";

// Use weight branch
TString cut_dy = TString::Format("(sist_mass > 0 && %s) * weight",
                                   proton_cut_mc.Data());
TString cut_ttbar = TString::Format("(sist_mass > 0 && %s) * weight",
                                     proton_cut_mc.Data());
```

```

// Fill histograms
t_dy->Draw(var + " >> h_dy", cut_dy, "goff");
t_ttbar->Draw(var + " >> h_ttbar", cut_ttbar, "goff");

// Apply scale factors
h_dy->Scale(DY_SCALE);
h_ttbar->Scale(TTBAR_SCALE);

```

10.2 Proton Cuts

Data/QCD (real protons):

```
TString proton_cut_data = "Sum$(proton_multi_arm==0)>0 && Sum$(proton_multi_arm==1)>0";
```

MC/Signal (pileup protons):

```
TString proton_cut_mc = "xi_arm1_1 >= 0 && xi_arm2_1 >= 0";
```

11. Reference Values Summary

Parameter	Value	Notes
Luminosity	54,900 pb ⁻¹	2018 UL
sigma(DY)	6077.22 pb	NNLO
sigma(ttbar)	831.76 pb	NNLO+NNLL
Sum_w(DY)	3.323×10^{12}	From NanoAOD
Sum_w(ttbar)	6.108×10^{11}	From NanoAOD
DY_SCALE	1.004×10	$L \times \text{sigma} / \text{Sum_w}$
TTBAR_SCALE	1.0	Pre-normalized
Proton acceptance	0.245	Both arms
Avg gen_weight (DY)	~17,000	aMC@NLO
Avg gen_weight (ttbar)	~2,000	aMC@NLO

12. File Locations

All files are stored on EOS:

```

/eos/user/m/mbianco/samples_2018_mutau/
  Data_2018_UL_MuTau_nano_merged_proton_vars.root
  QCD_2018_UL_MuTau_nano_merged_proton_vars_new.root
  DY_2018_UL_MuTau_nano_merged_pileup_protons.root
  ttjets_2018_UL_MuTau_nano_merged_pileup_protons.root
  ...

```

```
/eos/user/m/mblancco/tau_analysis/MuTau_channel/  
MuTau_sinal_SM_2018_july.root
```

13. Scripts Reference

Script	Purpose
<code>check_mc_normalization.py</code>	Calculate Sum_w from original NanoAOD
<code>check_weights_in_files.py</code>	Verify weight values in merged files
<code>validate_normalization.py</code>	Validate normalization approach
<code>MuTauPlots_after.C</code>	Main plotting macro (updated)
<code>fase1_dy.py</code>	DY phase 1 processing
<code>fase1_ttjets.py</code>	ttbar phase 1 processing
<code>merge_pp_mutau.py</code>	Merge and add proton mixing

14. Cross-Section References

- **DY (NNLO):** FEWZ 3.1, Phys.Rev.D 86 (2012) 094034
 - **ttbar (NNLO+NNLL):** TOP++ 2.0, Comput.Phys.Commun. 185 (2014) 2930
 - **CMS Standard Cross-sections:** <https://twiki.cern.ch/twiki/bin/viewauth/CMS/StandardModelCrossSections>
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15. Conclusion

The normalization is validated to be correct:

1. DY uses `weight × DY_SCALE` where $DY_SCALE = L \times \sigma / \text{Sum_w} = 1.004 \times 10$
2. ttbar uses `weight × 1.0` since the 0.15 factor already approximates $L \times \sigma / \text{Sum_w} \times \text{avg_gen_w}$
3. The 0.15 factor matches theoretical calculation to within 0.3%
4. Data/MC ratios in control regions should be ~ 1.0

The empirical 0.15 factor used for ttbar is not arbitrary - it is mathematically equivalent to the proper normalization formula.